Assignment Topic :

Programming_Fundamental_Lab _05

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Submitted To :

SIR ASAD.

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Task 1

Write a C++ program to take a number from user and print its multiplication table from 1 to 10.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int num;

    cout << "Enter a number: ";
    cin >> num;

    cout << "\nTable of " << num << ":\n";
    for (int i = 1; i <= 10; i++) {
        cout << num << " * " << i << " = " << num * i << endl;
    }

    return 0;
}
```

Output:

```
Enter a number: 5
Table of 5:
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

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lab 5.1.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;

    cout << "Enter a number: ";
    cin >> num;

    cout << "\nTable of " << num << ":\n";
    for (int i = 1; i <= 10; i++) {
        cout << num << " * " << i << " = " << num * i << endl;
    }

    return 0;
}
```

```
Enter a number: 5

Table of 5:
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50

--------------------------------
```

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Task 2

```
#include <iostream>
using namespace std;

int main() {
    int base, exponent;
    long long result = 1;

    cout << "Enter base: ";
    cin >> base;
    cout << "Enter exponent: ";
    cin >> exponent;

    for (int i = 1; i <= exponent; i++) {
        result = result * base;
    }

    cout << base << " ^ " << exponent << " = " << result << endl;

    return 0;
}
```

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```cpp
#include <iostream>
using namespace std;

int main() {
    int base, exponent;
    long long result = 1;

    cout << "Enter base: ";
    cin >> base;
    cout << "Enter exponent: ";
    cin >> exponent;

    for (int i = 1; i <= exponent; i++) {
        result = result * base;
    }

    cout << base << " ^ " << exponent << " = " << result << endl;

    return 0;
}
```

```
C:\Users\PMLS\OneDrive\Des
Enter base: 8
Enter exponent: 4
8 ^ 4 = 4096

--------------------------------
```

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Task 3

```
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;

    cout << "Enter a positive number: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        sum = sum + i;
    }

    cout << "Sum of natural numbers from 1 to " << n
         << " = " << sum << endl;

    return 0;
}
```

Output:

```
Enter a positive number: 2
Sum of natural numbers from 1 to 2 = 3

Enter a positive number: 5
Sum of natural numbers from 1 to 5 = 15
```

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5.1.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;

    cout << "Enter a positive number: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        sum = sum + i;
    }

    cout << "Sum of natural numbers from 1 to " << n << " = " << sum << endl;

    return 0;
}
```

```
C:\Users\PMLS\OneDrive\Des
Enter a positive number: 2
Sum of natural numbers from 1 to 2 = 3

--------------------------------
```

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Task 4

```
#include <iostream>
using namespace std;

int main() {
    int num, digit, sum = 0;

    cout << "Enter a number: ";
    cin >> num;

    while (num > 0) {
        digit = num % 10;    // get last digit
        sum = sum + digit; // add it to sum
        num = num / 10;    // remove last digit
    }

    cout << "Sum of digits = " << sum << endl;

    return 0;
}
```

Output :

```
Enter a number: 6
Sum of digits = 6
```

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```cpp
#include <iostream>
using namespace std;

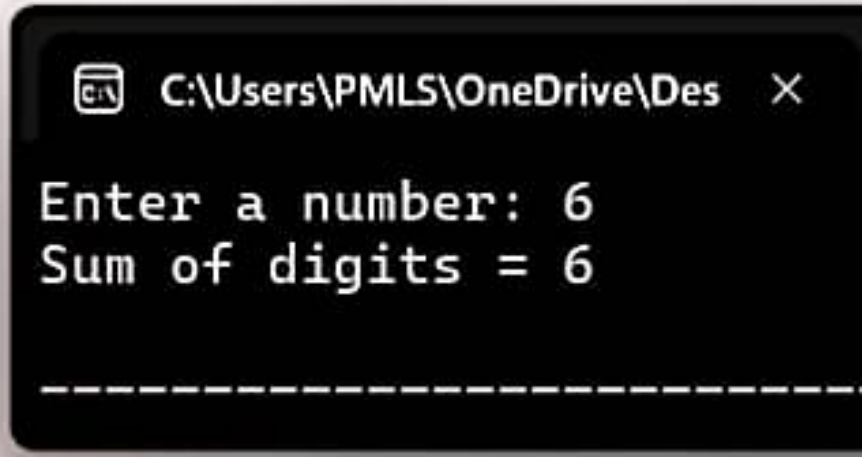
int main() {
    int num, digit, sum = 0;

    cout << "Enter a number: ";
    cin >> num;

    while (num > 0) {
        digit = num % 10;   // get last digit
        sum = sum + digit;  // add it to sum
        num = num / 10;     // remove last digit
    }4

    cout << "Sum of digits = " << sum << endl;

    return 0;
}
```



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Task 5

```
#include <iostream>
using namespace std;

int main() {
    int num;
    long long fact = 1;

    cout << "Enter a number: ";
    cin >> num;

    for (int i = 1; i <= num; i++) {
        fact = fact * i;
    }

    cout << "Factorial of " << num << " = " << fact << endl;

    return 0;
}
```

Output:

```
Enter a number: 4
Factorial of 4 = 24
```

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```cpp
#include <iostream>
using namespace std;

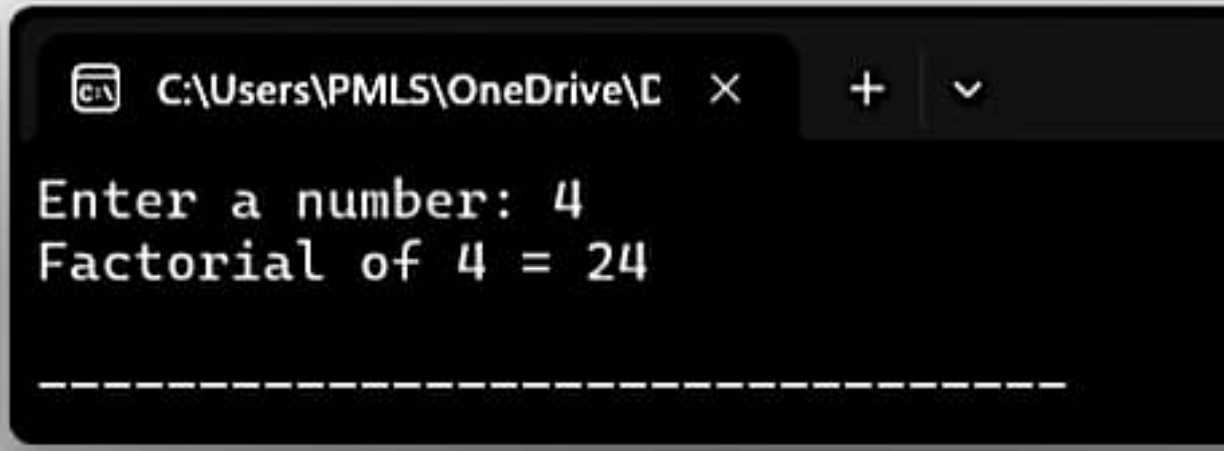
int main() {
    int num;
    long long fact = 1;

    cout << "Enter a number: ";
    cin >> num;

    for (int i = 1; i <= num; i++) {
        fact = fact * i;
    }

    cout << "Factorial of " << num << " = " << fact << endl;

    return 0;
}
```



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Task 7

```
#include <iostream>
using namespace std;

int main(){
    for (int i = 1; i <= 7; i++) {
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

Output:

```
*
**
***
****
*****
******
*******
```

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```cpp
#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 7; i++) {
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }

    return 0;
}
```

```
C:\Users\PMLS\OneDrive\Des
*
**
***
****
*****
******
*******

--------------------------------
Process exited after 0.3433 seconds with retu
Press any key to continue . . .
```

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Task 7 (Part a)

```
#include <iostream>
using namespace std;

int main()
{
   int month;

   cout << "Enter month number (1-12): ";
   cin >> month;

   switch (month) {
       case 1:
       case 3:
       case 5:
       case 7:
       case 8:
       case 10:
       case 12:
           cout << "31 days" << endl;
           break;
       case 4:
       case 6:
       case 9:
       case 11:
           cout << "30 days" << endl;
           break;
       case 2:
           cout << "28 or 29 days" << endl;
           break;

       default:
           cout << "Invalid month number" << endl;
   }

    return 0;
}
```

Output:

```
Enter month number (1-12): 9
30 days
```

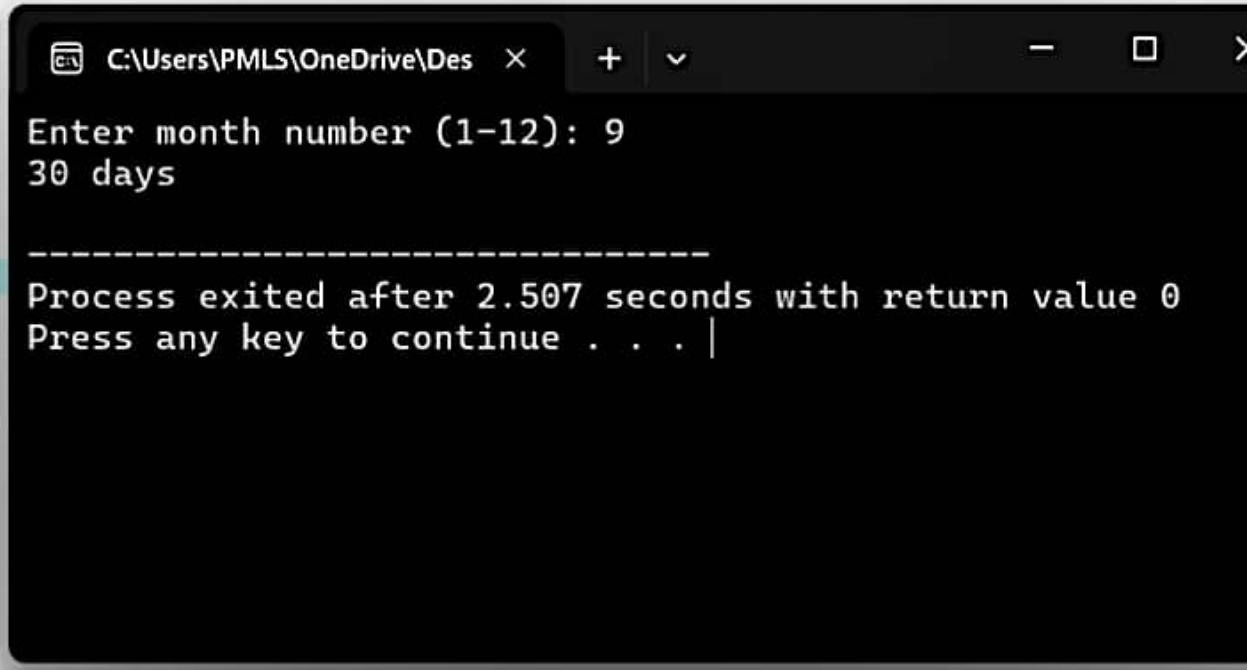
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```cpp
int main() {
    int month;

    cout << "Enter month number (1-12): ";
    cin >> month;

    switch (month) {
        case 1:
        case 3:
        case 5:
        case 7:
        case 8:
        case 10:
        case 12:
            cout << "31 days" << endl;
            break;
        case 4:
        case 6:
        case 9:
        case 11:
            cout << "30 days" << endl;
            break;
        case 2:
            cout << "28 or 29 days" << endl;
            break;
        default:
            cout << "Invalid month number" << endl;
    }

    return 0;
```



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Task 7 (Part b)

```
#include <iostream>
using namespace std;

int main()
{
    char ch;

    cout << "Enter an alphabet: ";
    cin >> ch;

    switch (ch) {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
            cout << ch << " is a vowel" << endl;
            break;
        default:
            if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z'))
                cout << ch << " is a consonant" << endl;
            else
                cout << "Not an alphabet" << endl;
    }

    return 0;
}
```

Output:

```
Enter an alphabet: a
a is a vowel

--------------------------------

Process exited after 2.405 seconds with return value 0
Press any key to continue . . .
```

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```cpp
#include <iostream>
using namespace std;

int main() {
    char ch;

    cout << "Enter an alphabet: ";
    cin >> ch;

    switch (ch) {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
        case 'A':
        case 'E':
        case 'I':
        case 'O':
        case 'U':
            cout << ch << " is a vowel" << endl;
            break;
        default:
            if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z'))
                cout << ch << " is a consonant" << endl;
            else
                cout << "Not an alphabet" << endl;
    }

    return 0;
}
```

```
C:\Users\PMLS\OneDrive\Des
Enter an alphabet: a
a is a vowel

--------------------------------
Process exited after 2.405 seconds with return value 0
Press any key to continue . . .
```

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